ARIANA CHIN

University of California 520 Portola Plaza Los Angeles, CA 90095 arianagchin@math.ucla.edu

RESEARCH INTERESTS: Zamolodchikov periodicity, cluster algebras, algebraic combinatorics
ADVISOR: Pavel Galashin

EDUCATION

University of California, Los Angeles

PhD in Mathematics Expected June 2027
PhD Candidate May 2025
M.A. in Mathematics March 2024
Dissertation Year Award - UCLA Department of Mathematics fellowship 2026-2027
Liggett Teaching Award - awarded to a teaching assistant for outstanding performance 2025-2026
Balbes Award - awarded to the top performing female graduate student 2023-2024
Teaching Experience:

- Upper Division: Probability and Statistics, Enumerative Combinatorics, Real Analysis
- Lower Division: Discrete Mathematics, Linear Algebra, Multivariable Calculus

University of California, Berkeley

B.A. in Mathematics, B.A. in Computer Science May 2022
Magna Cum Laude GPA: 3.96

- Percy Lionel Davis Award - awarded to a senior or seniors who have demonstrated excellent scholarship in mathematics
- Regents' and Chancellor's Scholar

Languages: Python, Java, Scheme, C, JavaScript, RISC-V, Matlab, LaTeX, SQL, MongoDB, DynamoDB

RESEARCH

Preprints and Publications:

"Homomesy of Tropical T-systems: Finite Type" [arXiv: 2610.02548] 2026
- Joint with Pavlo Pylyavskyy

"Half-periodicity of Zamolodchikov periodic cluster algebras" [arXiv: 2602.15140] 2026
- Submitted to *International Mathematics Research Notices (IMRN)*

"Classification of Zamolodchikov periodic cluster algebras" [arXiv: 2510.18031] 2025
- Submitted to *Advances of Mathematics*

"F-Polynomial Ratios in the r-Kronecker" [pdf][slides] 2021
- Joint with Noah Caplinger, Nyah Davis, and Swapnil Garg

"Lattice Models and Puzzles for Dual Weak Symmetric Grothendieck Polynomials" [pdf][slides] 2021
- Joint with Elisabeth Bullock, Noah Caplinger, Nyah Davis, and Gahl Shemy

"The Polynomial Learning With Errors Problem and the Smearing Condition" [paper] 2022
- Joint with Liljana Babinkostova, Aaron Kirtland, Esther Plotnick, and Vladyslav Nazarchuk
- *Journal of Mathematical Cryptology*, vol. 16, no. 1, 2022, pp. 215-232.

"The Ring Learning With Errors Problem: Spectral Distortion" [paper] 2025
- Joint with Liljana Babinkostova, Aaron Kirtland, Esther Plotnick, and Vladyslav Nazarchuk
- *Involve: a Journal of Mathematics*, vol. 18, no. 2, 2025, pp. 181-198.

Manuscripts in Preparation:

"Enumeration of Signed Seeds"
- Joint with A. Burcoff, M. Elkin, S. Kim, T. Martinez, A. Schwartz, and M. Tyler

CONFERENCES, WORKSHOPS & TALKS

- Combinatorics and Arithmetic for Physics (CAP) - Speaker — November 2026
  Institut des Hautes Études Scientifiques (IHES) — Bures-sur-Yvette, France
- AMS Sectional: Diagrammatics for Cluster Algebras and Physics - Speaker — November 2026
  Arizona State University — Tempe, AZ
- University of Michigan Combinatorics Seminar - Invited Speaker — October 2026
  University of Michigan, Ann Arbor — Ann Arbor, MI
- Richard P. Stanley Seminar in Combinatorics - Invited Speaker — October 2026
  Harvard University — Boston, MA
- IPAM Tutorials: Quantum Topology, Character Varieties - Lecturer — September 2026
  Institute for Pure and Applied Mathematics, UCLA — Los Angeles, CA
- Formal Power Series and Algebraic Combinatorics (FPSAC) - Presenter — July 2026
  University of Washington — Seattle, WA
- CMND 2026: Algebraic Combinatorics and Applications - Lightning talk — South Bend, IN
- Cornell Combinatorics Seminar (2026) - Invited Speaker — April 2026
  Cornell University — Ithaca, NY
- UIUC Combinatorics Seminar (2026) - Invited Speaker — April 2026
  University of Illinois, Urbana-Champaign — Urbana-Champaign, IL
- ALGECOM 2026 - Presenter — West Lafayette, IN
- UMN Combinatorics Seminar (2026) - Invited Speaker — April 2026
  University of Minnesota, Twin Cities — Minneapolis, MN
- NCUWM 2026 - Invited Panelist — Lincoln, NE
- Summer School on Algebraic Combinatorics (SSAC) 2025 — Seoul, Korea
- Formal Power Series and Algebraic Combinatorics (FPSAC) 2025 — Sapporo, Japan
- Graduate Student Combinatorics Conference (GSCC) 2025 - Presenter — Los Angeles, CA
- Formal Power Series and Algebraic Combinatorics (FPSAC) 2024 — Bochum, Germany

OTHER INTERESTS

UCLA Women in Mathematics - organizer — 2024-2026
UCLA Symphony, UCLA Philharmonia - violin — 2022-2026
UC Berkeley Symphony Orchestra - violin — 2019-2022
UC Berkeley Chamber Orchestra - violin — 2018-2019